

DIGITAL HARDWARE

→ Standard chips

→ Programmable Logic Devices

- PLA (Programmable Logic Array)

- PAL (—————)

- GAL (Generic Array Logic)

- CPLD's (Complex Programmable Logic Devices.)

- FPGA's (Field-Programmable Gate Array)

→ Custom-Designed Chips

intel →

ALFRA
Cyclone

AMD → XILINX.

ARTIX

Transistor $\rightarrow$ Logic Gate $\rightarrow$ Subcircuit
lvl $\rightarrow$ lvl $\rightarrow$ Level

CAD Tools

- Xilinx - Vivado
- Intel - Quartus Prime
- Mentor Graphics - ModelSim, Precision
- Microsemi - Libero

HDL (Hardware Descriptive Language)

- Verilog (System Verilog)
- VHDL (Very high speed integrated circuit hardware language)

$\hookrightarrow$ Synthesis : conv. of verilog test code to actual logic circuit

output : netlist

connection of 

↳ Xilinx Nomenclature

4 Logic Cells (LC) = Slice

2 slice = configurable Logic Block (CLB)

	<u>Slices</u>	<u>RAM (Kb)</u>	<u>FF</u>
Nexys A750T →	8150	600	65,200
—— A7100T →	15,850	1,188	126,800

* LUT (lookup tables = slices x 4)
FF = LUT x 2

① Behavioural Simulation :

- fastest one
- it is based on code & not on our synthesized

version

② Implementation

- It is mostly placement & routing

uses an numeric algorithm
to choose required no. of
logic cells out of total

→ upon cad tool

- * In implementation: Report Utilization
to see how much hardware is
being used

③ Generate Bitstream:

↳ If directly pressed, it goes through

synthesis, implementation then
comes to gen. bitstream

* open Hardware Manager in Program
& Debug & open target

↳ for board files : Digilent FPGA
Board files